

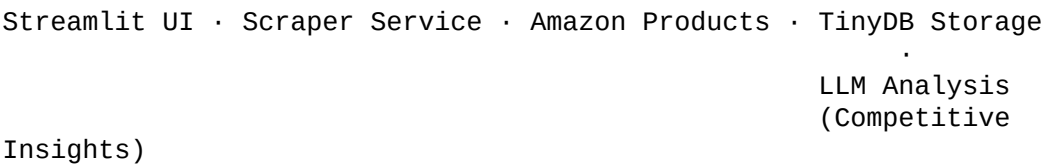
Amazon Price Competitor Analysis LLM

A **Streamlit dashboard** for scraping Amazon product data by ASIN, discovering competitors, and generating **LLM-powered competitive analysis** reports.

· Features

- **·ASIN Lookup** · Scrape product details by Amazon Standard Identification Number
 - **·Geo-targeting** · Localized results by country/region
 - **·Competitor Discovery** · Automatically find and analyze competing products
 - **·LLM Analysis** · AI-powered pricing and positioning insights
 - **·TinyDB Storage** · Lightweight local JSON-based persistence
 - **·Product Cards** · Visual display with images and pricing
-

.. Architecture



· Getting Started

```
cd AmazonPriceCompetitorAnalysisLLM
uv sync
uv run streamlit run main.py
```

Environment Variables (.env)

```
OPENAI_API_KEY=sk-...
BRIGHT_DATA_API_KEY=...
```

· Logic Flow

- 1 **Input** · User enters ASIN and target country
- 2 **Scrape** · Product data fetched from Amazon
- 3 **Store** · Data persisted to TinyDB
- 4 **Competitors** · Related products discovered and scraped
- 5 **Display** · Product cards rendered (10 per page)
- 6 **Analyze** · LLM generates competitive insights

Key Functions

Function Purpose	----- -----	<code>scrape_and_store_product()</code> Fetch Amazon product by ASIN	
<code>fetch_and_store_competitors()</code> Discover competing products		<code>analyze_competitors()</code>	

· Dependencies

langchain>=0.3.27, langchain-openai>=0.3.33, openai>=1.107.2, streamlit>=1.49.1,
tinydb>=4.8.2, python-dotenv>=1.1.1

· License

Educational project · use freely for learning and reference.